MILITARY SPECIFICATION

PROPELLANT, T36E1, FOR USE IN CHARGE, PROPELLING, 155MM, XM-115 (WHITE BAG)

1. SCOPE

1.1 This specification covers the T36E1 propellant for use in 155MM Propelling charges designated as XM-115.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-G-155 - Graphite (For Use in Ammunition)
MIL-P-193 - Potassium Sulfate (For Ordnance Use)
MIL-N-244 - Nitrocellulose (For Use in Ammunition)
MIL-N-246 - Nitroglycerin
MIL-E-255 - Ethyl Centralite (Carbamite)
JAN-N-494 - Nitroguanidine (Picrite)
MIL-A-2550 - Ammunition and Special Weapons; General Specification for
MIL-I-45208 - Inspection Requirements, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Inspection Terms and Definitions.
MIL-STD-286 - Propellants, Solid; Sampling, Examination and Testing.
MIL-STD-414 - Sampling, Procedures and Tables for Inspection by Variables for Percent Defective.

FSC: 1375

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DRAWINGS

ORDNANCE CORPS

76-4-46 - Box, Packing with Metal Liner, M24, for Smokeless Powder Assembly and Details
76-4-56 - Box, Packing, Metal-Wood M17 for Smokeless Propellant
8858848 - Marking Diagram and Sealing of Metal Lined Wooden Packing Boxes for Shipment of Propellant
119249 - Charge, Propelling, 155MM, XM115 (White Bag) for Howitzer T255 and T258

PUBLICATIONS

ORDNANCE CORPS

ORD-M608-11 - Procedures and Tables for Continuous Sampling by Attributes.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-The constituent materials shall comply with the requirements of the applicable specification as follows:

Constituent Material	Conforming to Specification
Nitrocellulose	MIL-N-244, Grade A, Type I
Nitroglycerin	MIL-N-246, Type I
Nitroguanidine	JAN-N-494, (Class to be as specified in contract, see 6.1)
Ethyl Centralite	MIL-E-255, Class 2
Graphite	MIL-G-155
Potassium Sulfate	MIL-P-193

3.2 Composition.-The composition, exclusive of total volatiles, shall be as specified in Table I, when determined as specified in the applicable sub-paragraphs of 4.3.1.

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Table I - Composition

Ingredient	Percent
Nitrocellulose	28.0 plus or minus 0.75
Nitroglycerin	22.5 plus or minus 0.75
Nitroguanidine	47.0 plus or minus 1.0
Ethyl Centralite	1.5 plus or minus 0.1
Potassium Sulfate	1.0 plus or minus 0.3
Graphite (glaze) added	0.1 maximum (max.)

3.3 Total volatiles.-The total volatiles content of the propellant shall be 0.50 percent max., when determined as specified in 4.3.2.

3.4 Heat test, 120 degree Centigrade (C).-The propellant shall not cause complete fading of the methyl violet test paper to a salmon-pink color in less than 40 minutes, when tested as specified in 4.3.3.

3.5 Form and dimensions of grain.

3.5.1 Form.-The grain shall be cylindrical with 7 longitudinal perforations. One in the center of the grain and six at the vertices of a symmetrical hexagon.

3.5.2 Length-diameter ratio.-The average grain length (L) shall be from 2.10 to 2.50 times the average grain diameter (D).

3.5.3 Length and diameter uniformity.-The length and diameters of grain shall comply with the uniformity requirements as shown in Table II.

Table II

Dimensions	Max.
L	6.25
D (grains 0.2 inch or more in diameter)	3.125
D (grains less than 0.2 inch in diameter)	6.25

3.5.4 Grain diameter-Perforation diameter ratio.-The average grain diameter (D) shall be not less than five nor more than 15 times the average diameter of the perforation (d).

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3.5.5 Web measurements.-The difference between the average outer web-thickness (Wo) and the average inner web thickness (Wi) shall not exceed 15 percent of the average web thickness (Wa).

3.6 Ballistics.

3.6.1 Velocity and pressure.-The propellant shall be assessable to impart zone velocities and pressures in accordance with Table III.

Table III

Charge	Velocity meters per second (mps)	Dynamic Piezo Electric Pressure Pounds per square inch (psi)	
		min.	max.
1-4	282	8,200	9,000
5	358	12,400	—
6	493	23,100	—
7	655	45,000	50,400

3.6.2 The lot standard diviation in velocity shall not exceed that shown in Table IV.

Table IV

Charge	Standard Deviation (mps)
4	1.5
6	1.8
7	2.0

3.6.3 Individual maximum pressure.-The pressure obtained from any excess charge round or any service condition shall not exceed 115 percent of the rated max. pressure. The excess charge is defined as 105 percent of the zone 7 charge weight.

3.7 Workmanship.-The best commercial practices shall be used in the formulation of propellant furnished under this specification, and all other

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applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL-I-45208.

4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.
b. Number of units of product inspected.
c. Results obtained for all inspection performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of analysis on all material procured directly by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.
h. Eight copies of Propellant Description Sheet 00 Form 1204, Containing the applicable data.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.1.3 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL-I-45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions.

4.2.1 Lot formation.-A lot shall consist of one or more batches of T36E1 Propellant produced by one manufacturer, in accordance with the same specification or same specification revision, under one continuous set of operating conditions. Each batch shall consist of that quantity of T36E1 Propellant which has been subjected to the same unit chemical or physical mixing process intended to make the final product homogenous.

4.2.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Handbook ORD-M608-11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect a nd an AQL of 0.40 percent for each minor defect.

4.2.2.1 Box or container, prior to filling (see drawing (dwg.) 76-4-46 or 76-4-56)

Categories	Defects	Method of Inspection	Code No (see 6.5)
Critical: None defined			
Major:	AQL 0.25 percent		
101.	Cracks or hole in metal liner (as applicable)	Visual	01001
102.	Foreign matter on interior	Visual	01002
Minor: None defined			

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4.2.2.2 Sealed Box or Container (see dwg. 76-4-46, or 76-4-56).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Cover incompletely closed	Visual-Manual	02001
102.	Gasket missing or broken	Visual	02002
103.	Wood cover split or otherwise damaged (when applicable)	Visual	02003
Minor:	AQL 2.50 percent		
201.	Car seal missing, unsealed or improperly positioned	Visual-Manual	02004
202.	Crack or hole in metal container	Visual	02005
203.	Hardware improperly engaged	Visual	02006
204.	Marking misleading or unidentifiable	Visual	02007
205.	Bare areas in exterior coating of metal container the sum of which is in excess of 1/4 square inch	Visual	02008

4.2.3 Testing.

4.2.3.1 Chemical composition, volatiles and stability, (see 3.2, 3.3, and 3.4) - Major defects - Six pounds of propellant shall be selected at random from the lot. One pound shall be used for chemical, volatiles and stability tests. If the sample fails to comply with the requirements specified the lot shall be rejected.

	Chemical Composition	Code No
a.		
	1. Nitrocellulose	03001
	2. Nitroglycerin	03002
	3. Nitroguanidine	03003
	4. Ethyl Centralite	03004
	5. Potassium Sulfate	03005
	6. Graphite	03006
b.	Total volatiles	04001
c.	Heat test	05001

The remaining five pounds shall be forwarded to the Commanding Officer, Picatinny Arsenal, Dover, New Jersey, for 65.5 degree Centigrade surveillance tests. All samples shall be packed in air tight containers and shall be marked to show the

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propellant designation, lot number, manufacturer, date of sampling, contract number and number of pounds in the lot (see 6.4).

4.2.3.2 Form and dimension (see 3.5) Major defect.-Thirty grains shall be selected from each lot for these examinations. If the sample fails to meet the requirements specified the lot shall be rejected.

	Type I Code No.
a. Form	06001
b. Length-diameter ratio	06002
c. Length-diameter uniformity	06003
d. Grain diameter-perforation diameter ratio	06004
e. Web measurements	06005

4.2.3.3 Six hundred pounds of propellant shall be randomly selected from the lot for the ballistics test.

For the first three lots of propellant charge from any production run, a 7 round uniformity series shall be fired for information at 140 degrees Fahrenheit (F.) and minus 40 degrees F. in addition to firings required to determine compliance with 3.6 at 70 degrees F.

4.2.3.3.1 The lot of propellant shall be rejected if not capable of assessment to the requirements of 3.6.1 and 3.6.2.

a. Velocity	Code No
Charge 4	07001
Charge 6	07002
Charge 7	07003
b. Pressure	Code No
Charge 4	08001
Charge 6	08002
Charge 7	08003

4.2.3.3.2 Uniformity of velocity (Major defect).-The lot of propellant shall be rejected if the sample standard deviation (see 6.2) at any zone tested (see Table III, when multiplied by the factor 0.69, exceeds the applicable requirements Code No 09001.

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4.2.3.3.3 Individual maximum pressure.-Critical defect - The lot of propellant shall be rejected if any individual round fails to comply with the applicable requirement Code No 10001.

4.2.3.4 Retest.-If for any reason the proof director considers that the test conditions have detrimentally affected the test resulst, he shall declare the results invalid and perform a new test on additional charges as required.

4.2.3.5 Air pressure tests (see dwg. 8858848) Major defect.-The containers shall be checked 100 percent as specified in 4.3.6. If any container fails to meet the requirement it shall be rejected and removed from the lot.

4.3 Test methods and procedures.

4.3.1 Composition.

4.3.1.1 Nitroglycerin.-The percentage of nitroglycerin shall be determined in accordance with Standard MIL-STD-286, Method 208.1 using pentane-methylene chloride azeotrope (two volumes of technical grade pentane to one volume of methylene chloride) as the solvent for extraction, except that the nitroglycerine content shall be calculated on a graphite (glaze) and total volatiles-free basis, and a tall form sintered glass filtering crucible shall be used to hold the specimen during the extraction. The crucible and contents shall be saved for the determination of nitroguandidine.

4.3.1.2 Nitroguanidine.-The percentage of nitroguanidine shall be determined in accordance with Standard MIL-STD-286, Method for extraction, except using the crucible and contents from 4.3.1.1 and that the nitroguanidine content shall be calculated on a graphite (glaze) and total volatile free basis.

4.3.1.3 Ethyl centralite.-The percentage of ethyl centralite shall be determined as specified in Standard MIL-STD-286, method 202.2 using pentane-methylene chloride azeotrope as the solvent for extraction, except that the ethyl centralite content shall be calculated on a graphite (glaze) and total volatiles free basis.

4.3.1.4 Potassium sulfate.-The potassium sulfate content of the propellant shall be determined in accordance with Standard MIL-STD-286, method 310.4 except that the potassium sulfate content shall be calculated on a graphite (glaze) and total volatiles free basis.

4.3.1.5 Graphite.-The percentage of graphite shall be determined in accordance with Standard MIL-STD-286, method 308.1.

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4.3.1.6 Nitrocellulose.-The percentage of nitrocellulose shall be determined in accordance with Standard MIL-STD-286, method 209.2.

4.3.2 Total volatiles.-The percentage of total volatiles shall be determined in accordance with Standard MIL-STD-286, method 103.3.

4.3.3 Heat test, 120 degrees C.-The 120 degrees C. heat test shall be conducted in accordance with Standard MIL-STD-286, method 404.1.

4.3.4 Form and dimensions of grains.-Examination for form shall be visual; measurements and calculations for dimensions shall be conducted in accordance with Standard MIL-STD-286 method 504.1.

4.3.5 Ballistics.-Unless otherwise specified herein, the test shall be performed in accordance with applicable provisions of the Ordnance Proof Manual. Increment Charge weights shall be determined by firing propelling charges assembled in accordance with applicable drawing 119249 (or closed bomb firings) to meet the requirements specified in Table III and Table IV.

4.3.5.1 Uniformity series.-The propellant shall be conditioned to 70 degrees F. plus or minus 2 degrees for at least 16 hours and shall be fired within a minimum time after removal from the conditioning chamber. A seven round uniformity series shall be fired at each zone specified in Table IV with propelling charge assembled in accordance with dwg. 8863416. The mean velocity, velocity standard deviation, and mean of the maximum pressure for each seven round sample shall be calculated.

4.3.6 Air pressure test for containers.-The container shall be subjected to an internal pressure of 1/2 to 1 pound per square inch by a method satisfactory to the Contracting Officer's representative. A water manometer shall be assembled in the system. A drop of 0.7 inches or more on the manometer in 15 seconds shall be cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 Level B.-The propellant shall be packed in a container complying with dwg. 76-4-46 or 76-4-56. Containers shall be closed, sealed, tested and marked in accordance with the requirements of dwg. 8858848.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the following:

a. Title, number and date of this specification.

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b. Class of Nitroguanidine required (see 3.1).

c. Process.-Details of the manufacturing process and the equipment used by the contractor will be submitted to the procuring activity in writing prior to commencement of manufacture. Any deviation from this manufacturing process must be submitted in writing to the procuring activity prior to being put into effect.

6.2 Standard deviation.-The Standard deviation will be calculated with (N-1) as the divisor in a standard statistical technique equivalent to that shown in Standard MIL-STD-414, section B example B-1.

6.3 Submission of test data.-In addition to the normal distribution of records, when the propellant is procured by the Department of the Army, one copy of all ballistic test data will be forwarded to the following:

Commanding Officer
Picatinny Arsenal
Dover, New Jersey
ATTN: SMUPA-ND4

6.4 Surveillance samples.-The expense of surveillance testing shall be borne by the Government. All samples shall be packed in air-tight containers and shall be marked to show the propellant designation, the projectile for which manufactured, lot number, manufacturer, date of sampling, contract number and number of pounds in the lot.

6.5 Inspection code numbers.-The five-digit code numbers assigned to the inspection herein are to facilitate future data collection and analysis by the Government.

Notice.-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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